GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE

REHABILITATE NORTH RIM INFRASTRUCTURE

VOLUME 11 POWER DISTRIBUTION – ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	110G001	GENERAL	SHEET INDEX
2	110G002	GENERAL	GENERAL NOTES
3	111C101	POWER DISTRIBUTION	CIVIL SITE MAP
4	111C102	POWER DISTRIBUTION	CABLE ROUTE PLAN & PROFILE RSPH STA 1110+00 TO 1120+00
5	111C103	POWER DISTRIBUTION	CABLE ROUTE PLAN & PROFILE RSPH STA 1120+00 TO 1132+00
6	111C104	POWER DISTRIBUTION	CABLE ROUTE PLAN & PROFILE RSPH STA 1132+00 TO 1142+81
7	111C501	POWER DISTRIBUTION	INSTALLATION DETAILS UNDERGROUND
8	111E501	POWER DISTRIBUTION	VAULT AND SPLICE DETAILS

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO 75014 TOU VANG Date Signed

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO 41809 RYAN RHOADES ARIZONA, U.S.A. Date Signed

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. WALCHUK CADD N. LYNCH TECH. REVIEW: M. GOUGH DATE: AUG 2026	SUB SHEET NO. 110 G001	TITLE OF SHEET GENERAL SHEET INDEX NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 1 OF 8

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GENERAL

G1. SCOPE
THE NOTES ON THIS SHEET AND THE STANDARD STRUCTURAL DETAILS ARE GENERAL AND APPLY TO THE ENTIRE PROJECT WHETHER SPECIFICALLY CALLED OUT OR NOT, EXCEPT WHERE THERE ARE SPECIFIC INDICATIONS TO THE CONTRARY ON STRUCTURAL SHEETS. IF THERE ARE QUESTIONS, THEY SHALL BE SUBMITTED TO THE CONTRACTING OFFICER AND ANSWERED IN WRITING PRIOR TO CONSTRUCTION.

G2. APPLICABLE SPECIFICATIONS AND CODES
- A. INTERNATIONAL BUILDING CODE, IBC 2024 WITH APPLICABLE EDITIONS OF THE CODE REFERENCED STANDARDS.
- B. AUTHORITY HAVING JURISTICTION: NATIONAL PARK SERVICE (WIND, SNOW, AND SEISMIC DESIGN CRITERIA IN ACCORDANCE WITH THE MORE CONSERVATIVE OF COCONINO COUNTY, AZ AND NATIONAL PARK SERVICE).
- C. BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE, ACI 318-19
- D. ASTM MATERIAL STANDARDS AS NOTED

G3. DESIGN CRITERIA
1. APPLIES TO ALL STRUCTURES (UNO)

A. DEAD LOAD:
1. ACTUAL TRIBUTARY STRUCTURE WEIGHT

B. LIVE LOAD:

1. SLAB ON GRADE:	250 PSF
2. WALKWAYS, STAIRS, GRATING (UNO):	100 PSF

C. WIND:

1. ULTIMATE WIND SPEED (V_{ULT}):	111 MPH
2. TORNADO SPEED (V_T):	NOT APPLICABLE PER ASCE 7, FIG. 32.1-1 & 1-2
3. NOMINAL WIND SPEED (V_{ASD}):	86 MPH
4. RISK CATEGORY:	IV
5. EFFECTIVE AREA (A_E):	NOT APPLICABLE
6. EXPOSURE CATEGORY:	C
7. ALL STRUCTURES:	ENCLOSED
8. INTERNAL PRESSURE COEFFICIENT (GC_{pi}):	+/- 0.18
9. DESIGN WIND PRESSURE:	23 PSF
10. COMPONENTS AND CLADDING PRESSURE:	******

D. SEISMIC:
1. ABOVE GRADE, NON WATER BEARING STRUCTURES:

a. RISK CATEGORY:	IV
b. IMPORTANCE FACTOR:	1.5
c. SPECTRAL RESPONSE ACCELERATION, S_S =	0.420g
d. SPECTRAL RESPONSE ACCELERATION, S_1 =	0.110g
e. SITE CLASS:	C
f. SEISMIC DESIGN CATEGORY:	C
g. SPECTRAL RESPONSE ACCELERATION, S_{DS} =	0.31g
h. SPECTRAL RESPONSE ACCELERATION, S_{D1} =	0.11g
i. BASIC SEISMIC FORCE RESISTING SYSTEM:	ASCE 7-22, CHAPT 13: MECH EQUIPMENT SUPPORT STRUCTURES AND PLATFORMS, OTHER STRUCTURES.
j. ANALYSIS PROCEDURE:	EQUIVALENT LATERAL FORCE
k. COMPONENT RESONANCE DUCTILITY FACTOR:	1.0
l. COMPONENT STRENGTH FACTOR:	1.50
m. DESIGN BASE SHEAR:	14 LBS

E. SNOW LOAD:

1. GROUND SNOW LOAD (p_g)	=	130 PSF (INNER CANYON)
2. FLAT ROOF SNOW LOAD (p_f)	=	71 PSF
3. EXPOSURE FACTOR (C_e)	=	0.9
4. RISK CATEGORY	=	IV
5. THERMAL FACTOR (C_t)	=	1.1
6. SLOPED ROOF FACTOR (C_S)	=	0.91
7. WINTER WIND PARAMETER (W_2)	=	0.35

G4. THE FOLLOWING NON-CONTRACTUAL GEOTECHNICAL REPORT WAS DEVELOPED FOR THIS PROJECT AND IS THE BASIS OF THIS STRUCTURAL DESIGN:

GEOTECHNICAL FIRM NAME:	LITHOS ENGINEERING
ADDRESS:	2750 S. WADSWORTH BLVD, SUITE D-200 DENVER, CO 80227
REPORT NUMBER:	21186
REPORT DATE:	JANUARY 16, 2024

1. FROST DEPTH:		36 IN
2. NET ALLOWABLE BEARING PRESSURE (GRAVITY)	=	3 KSF
3. NET ALLOWABLE BEARING PRESSURE (WIND/SEISMIC)	=	4 KSF
4. COEFFICIENT OF FRICTION:	=	0.35
5. SUBGRADE MODULUS:	=	125 PSI/IN
6. ACTIVE LATERAL SOIL PRESSURE:	=	42 PCF
7. AT-REST LATERAL SOIL PRESSURE:	=	63 PCF
8. PASSIVE LATERAL SOIL PRESSURE:	=	200 PCF

G5. SAFETY
SAFETY AND STRUCTURE STABILITY DURING CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. STRUCTURES HAVE BEEN DESIGNED TO RESIST THE DESIGN LIVE LOADS ONLY AS A COMPLETED STRUCTURE.

GENERAL (CONTINUED)

G6. OPENINGS
OPENINGS FOR PIPES, DUCTS, CONDUITS, ETC. ARE NOT ALL SHOWN ON THE STRUCTURAL DRAWINGS. COORDINATE AND PROVIDE OPENINGS AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

G7. SPECIAL INSPECTIONS
SPECIAL INSPECTIONS ARE REQUIRED IN ACCORDANCE WITH CHAPTER 1 AND CHAPTER 17 OF THE IBC. THE CONTRACTOR IS RESPONSIBLE FOR STRUCTURAL SPECIAL INSPECTIONS (SSI) BY QUALIFIED PERSONNEL. THE CONTRACTOR SHALL PROVIDE FOR FULL ACCESS TO THE WORK BY STSI AGENTS AND SHALL PROVIDE FOR THESE INSPECTIONS IN THEIR CONSTRUCTION SCHEDULE IN ACCORDANCE WITH THE SPECIFICATIONS. THE SPECIAL INSPECTION PROGRAM AND STATEMENT OF SPECIAL INSPECTIONS SHALL BE IN ACCORDANCE WITH 01 40 00.

G8. GENERAL DETAILS
THE STANDARD DETAILS ON SHEET S200 DEPICT TYPICAL DETAILING TO BE USED ON THIS PROJECT. IF CONDITIONS ARE NOT EXPLICITLY SHOWN ON THE DRAWINGS THEY SHALL BE MADE SIMILAR TO THE STANDARD DETAILS. OBTAIN APPROVAL OF THE CONTRACTING OFFICER IN WRITING FOR SIMILAR CONDITIONS PRIOR TO CONSTRUCTION.

G9. FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL.

G10. SUBMIT FOR REVIEW ALL EQUIPMENT SIZES, OPERATING WEIGHTS, VIBRATION FORCES, SUPPORT LOCATIONS, ALONG WITH ANY FLOOR OPENINGS, NOTCHES, AND RECESSES REQUIRED BY SUCH EQUIPMENT. CONCRETE SUPPORT PADS AND/OR FRAMING REQUIRED TO SUPPORT SAID EQUIPMENT SHALL NOT BE FABRICATED AND PLACED UNTIL THE CONCRETE SUPPORT PADS AND/OR FRAMING IS APPROVED TO SUPPORT THE EQUIPMENT.

CONCRETE

C1. DESIGN STRENGTHS:
F'c = 4,500 PSI
REBAR Fy = 60,000 PSI

C2. CONCRETE COVER
UNLESS OTHERWISE NOTED, PROVIDE CONCRETE COVER FOR REINFORCING AS FOLLOWS:
CONCRETE DEPOSITED AGAINST EARTH: 3 INCH
FORMED SURFACES EXPOSED TO WEATHER, WATER, OR IN CONTACT WITH EARTH:
- a. 2 INCH

INTERIOR DRY SURFACES:
- a. 1-1/2 INCH FOR BEAMS, GIRDERS, AND COLUMNS.
- b. 1-1/2 INCH FOR SLABS, WALLS

ALL OTHER: 2 INCH
SEE DRAWINGS FOR EXCEPTIONS

C3. SEE SPECIFICATIONS FOR REINFORCING PLACEMENT REQUIREMENTS.

C4. REFER TO OTHER DISCIPLINE DRAWINGS PRIOR TO CONSTRUCTION FOR EMBEDDED ITEMS AND PENETRATIONS NOT SHOWN ON STRUCTURAL DRAWINGS. AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

C5. PROVIDE 3/4" CHAMFERS AT ALL EXPOSED EDGES. NOT ALL CHAMFERS MAY BE SHOWN ON DRAWINGS.

C6. FIELD ADJUST REINFORCING AT OPENINGS AND EMBEDDED ITEMS AS INDICATED.

C7. ANCHOR BOLTS NOT SPECIFIED BY CONTRACTING OFFICER SHALL BE DESIGNED AND CERTIFIED BY A REGISTERED PROFESSIONAL CONTRACTING OFFICER LICENSED IN THE STATE OF ARIZONA, RETAINED BY THE CONTRACTOR, IN ACCORDANCE WITH APPLICABLE PROJECT AND CODE REQUIREMENTS. SUBMIT AS A SHOP DRAWING FOR REVIEW AND APPROVAL BY THE CONTRACTING OFFICER. COORDINATE LOCATION, SIZE AND EMBEDMENT PRIOR TO CASTING CONCRETE.

C8. CONTINUOUS WATERSTOP SHALL BE INSTALLED IN JOINTS SUBJECT TO STATIC WATER PRESSURE.

C9. ABSOLUTELY NO WELDING OF REINFORCING BARS OR TORCHING TO BEND REINFORCING BARS SHALL BE ALLOWED WITHOUT SPECIFIC APPROVAL FROM THE CONTRACTING OFFICER.

C10. CONTRACTOR SHALL SUBMIT A CONCRETE PLACEMENT PLAN PER SPECIFICATION 03 31 31 IDENTIFYING JOINT TYPES, JOINT LOCATIONS AND CONCRETE PLACEMENT SEQUENCE.

C11. ALL CAST IN PLACE AND POST-INSTALLED ANCHORS INDICATED IN THE STRUCTURAL DOCUMENTS SHALL COMPLY WITH CHAPTER 17 OF ACI 318 AND CHAPTER 19 OF THE IBC. ALL EXPANSION AND ADHESIVE ANCHORS SHALL HAVE THE ICC REPORT SHOWING EQUIVALENT LOAD CAPACITY. SUBMIT AND INSTALL PER THE ICC EVALUATION REPORT.

DELEGATED SUBMITALS

D1. DEFINITION:
DELEGATED DESIGN SUBMITTALS ARE ITEMS FOR WHICH THE FINAL DESIGN OF A SPECIFIC ELEMENT OF THE PERMANENT CONSTRUCTION WORK EXPRESSLY DELEGATED TO A PROPERLY LICENSED AND REGISTERED DESIGN PROFESSIONAL RETAINED OR EMPLOYED BY THE CONTRACTOR, SUBCONTRACTOR, SUPPLIER OF ANY TIER. "DELEGATED DESIGN" APPLIES ONLY TO PERMANENT WORK THAT WILL BE PART OF THE COMPLETED PROJECT AS A FUNCTIONING WHOLE.

D2. DELEGATED DESIGN LIST:
THE FOLLOWING IS A LIST OF DELEGATED DESIGN SUBMITTALS THAT ARE EXPECTED TO CONTAIN STRUCTURAL CALCULATIONS OF SAFETY-RELATED SYSTEM INFORMATION FOR REVIEW. THESE SUBMITTALS APPLY TO DELEGATED DESIGNED SYSTEMS PRIOR TO INSTALLATION OF THE INDICATED STRUCTURAL ELEMENT, EQUIPMENT, DISTRIBUTION SYSTEM, COMPONENT, OR ITS ANCHORAGE. THE CONTRACTOR SHALL SUBMIT THE REQUIRED ENGINEER CERTIFICATION, SUPPORTING DATA, AND DRAWINGS FOR REVIEW AND ACCEPTANCE BY THE CONTRACTING OFFICER.

D3. DELEGATED DESIGN SUBMITTAL INCLUDE, BUT ARE NOT LIMITED TO, THE FOLLOWING:
1. MODULAR FRAMING SYSTEMS
2. EQUIPMENT ANCHORAGE AND BRACING WHERE REQUIRED BY TECHNICAL SPECIFICATIONS.

EARTHWORK

E1. ALL EARTHWORK AND SITE PREPARATION SHALL BE PERFORMED IN STRICT ACCORDANCE WITH THE SPECIFICATIONS. ALL FOUNDATION EXCAVATIONS SHALL BE OBSERVED AND APPROVED BY THE GEOTECHNICAL ENGINEER.

E2. ANY EXISTING FILLS OR UNSUITABLE SOILS AS DETERMINED BY THE GEOTECHNICAL ENGINEER SHALL BE EXCAVATED AND REPLACED WITH PROPERLY COMPACTED SELECT FILL PER SPECIFICATIONS.

E3. ALL FOUNDATIONS SHALL BEAR ON A MINIMUM DEPTH OF PROPERLY COMPACTED SELECT FILL AS INDICATED ON DRAWINGS.

E4. CONTRACTOR SHALL EXERCISE EXTREME CARE WHEN EXCAVATING OR GRADING ADJACENT TO EXISTING STRUCTURES SO AS TO NOT DAMAGE OR UNDERMINE FOUNDATIONS, WALLS, SLABS, OR UTILITIES, ETC.

E5. DO NOT EXCAVATE BELOW THE BEARING ELEVATION OF ANY COMPLETED FOOTING NOR ANY CLOSER TO THE FOOTING THAN A SLOPE OF TWO HORIZONTAL (MEASURED FROM EDGE OF FOOTING TO NEAREST POINT IN EXCAVATION) TO ONE VERTICAL.

STATEMENT OF STRUCTURAL TESTS AND SPECIAL INSPECTIONS

S1. REFER TO PROJECT MANUAL SECTION 01 40 00 - QUALITY REQUIREMENTS, AND SECTION 01 40 00-A - THE STATEMENT OF STRUCTURAL TESTS AND SPECIAL INSPECTIONS

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 85509 JON CORONADO Date Signed 04/17/2026 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. McCALLAN CADD R. NELSON TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 110 G002	TITLE OF SHEET POWER DISTRIBUTION GENERAL NOTES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 8

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200 0 200 400

SCALE OF FEET
1" = 200'

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE No. 75014 TOU VANG State Signed

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. WALCHUK DRAWN: N. LYNCH TECH. REVIEW: S. EVERETT DATE: MAY 2026	SUB SHEET NO. 111 C101	TITLE OF SHEET POWER DISTRIBUTION CIVIL SITE MAP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 8

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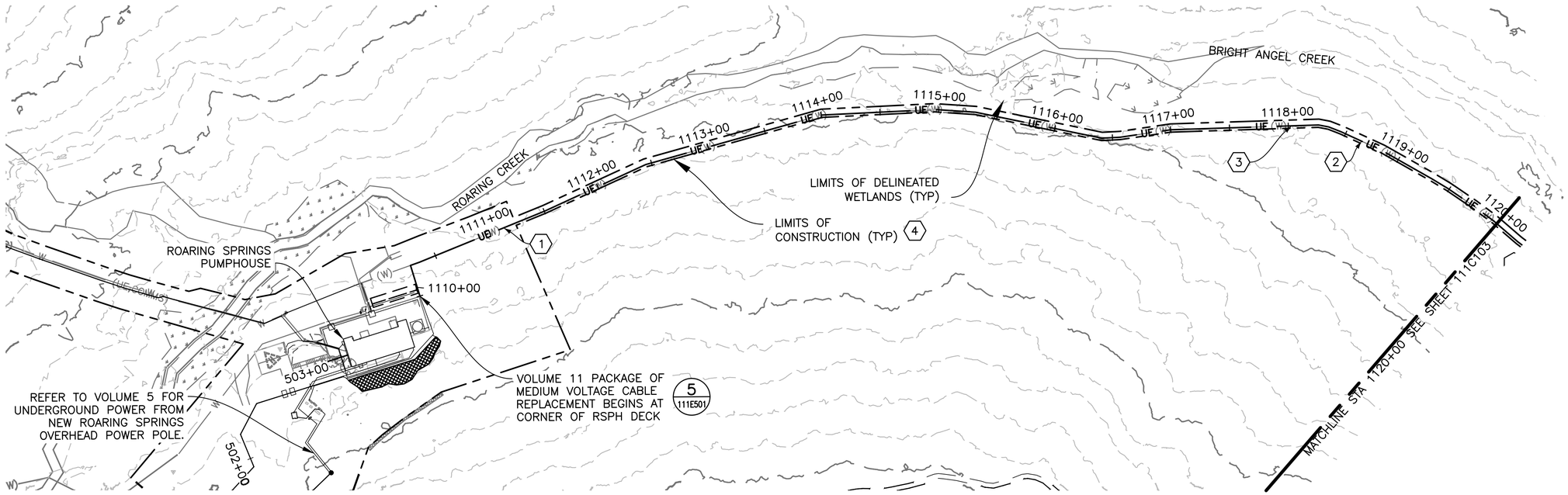

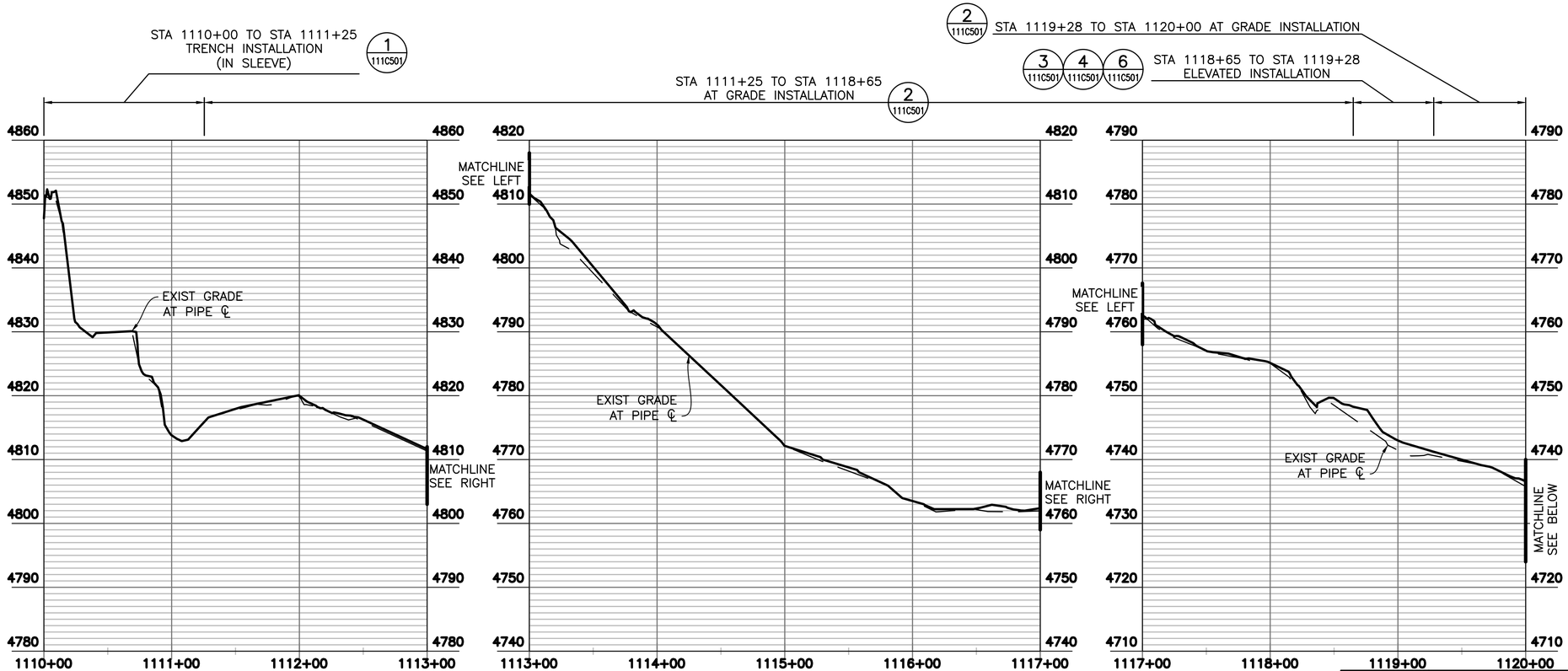

GENERAL NOTES:

A. EXIST MV CABLE TO BE REMOVED AND DISPOSED WHERE ACCESSIBLE WITHOUT EXCAVATION. ALL BURIED CABLE (SOIL OR ROCKFALL) TO BE CUT AS CLOSE AS POSSIBLE TO BURIED AREA AND ABANDONED IN PLACE.

B. CONTRACTOR TO RESTORE DISTURBED AREAS TO PRECONSTRUCTION CONDITIONS.

C. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY. GROUND DATA IS SHOWN TO PROVIDE A VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

D. EXIST ROARING SPRINGS PUMPHOUSE TO MANZANITA ALIGNMENT IS APPROXIMATE. MINOR DEVIATIONS IN THE NEW UTILITY ALIGNMENT ARE ANTICIPATED TO FACILITATE CONSTRUCTION.

E. CONTRACTOR TO USE 6–INCH HDPE PIPE AS–NEEDED FOR SACRIFICIAL EXTERIOR SLEEVE DURING MV CABLE INSTALLATION TO PREVENT CABLE INSULATION DAMAGE DURING INSTALL.

KEY NOTES: (X)

1. BURIED INSTALLATION UNTIL ALIGNMENT IS LOCATED IN THE TRAIL. ONCE IN TRAIL, ALIGNMENT IS AT GRADE INSTALLATION.

2. DRAINAGE CROSSINGS. PLACE MV CABLE INTO RIGID CONDUIT AND PLACE ADJACENT TO WATERLINE ON EXIST ROCK AND METAL PEDESTALS PER DETAILS 3 & 4 SHEET C501. EXTEND RIGID CONDUIT AT LEAST 20 LF ONTO AT GRADE PLACEMENT.

3. MV CABLE SURFACE SPLICE PER DETAIL 3 SHEET 111E501.

4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
1110+00	1111+25	SEE VOLUME 5 LIMITS OF CONSTRUCTION	
1111+25	1120+00	5 FT	2 FT

50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 75014 TOU VANG State of Arizona U.S.A.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 10-9-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	C. WALCHUK	111 C102	POWER DISTRIBUTION PLAN AND PROFILE STA 1110+00 TO 1120+00
	DRAFT N. LYNCH		
	TECH. REVIEW: S. EVERETT		
	DATE: JUNE 2026		NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	4 OF 8

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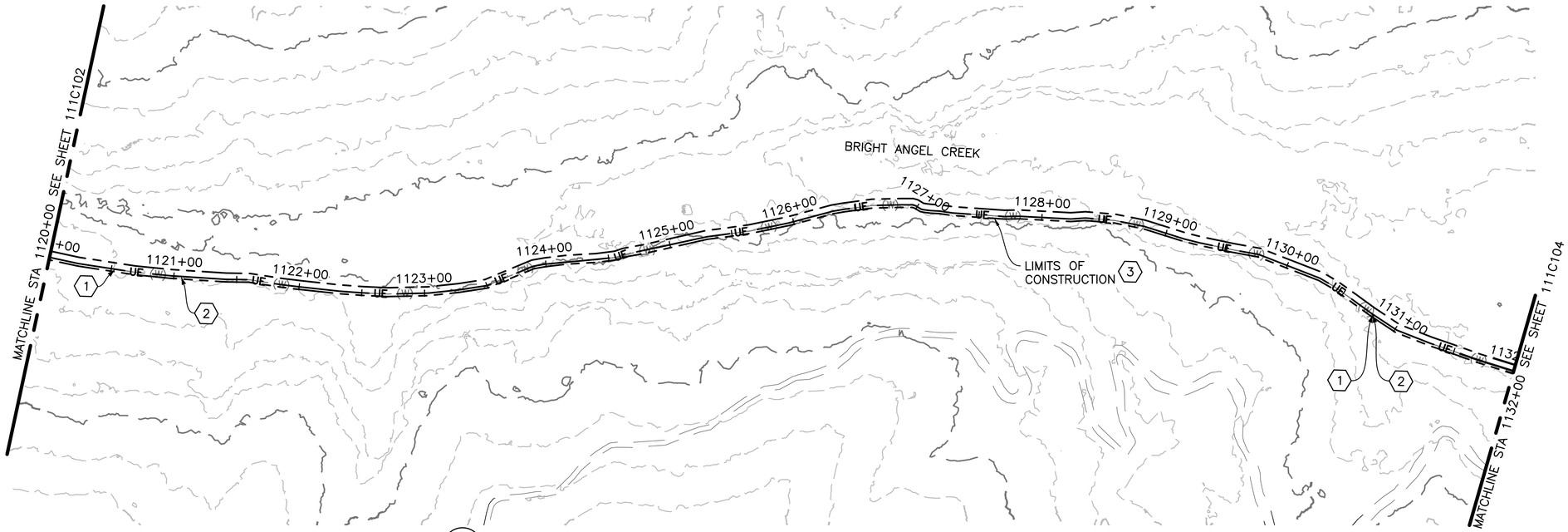


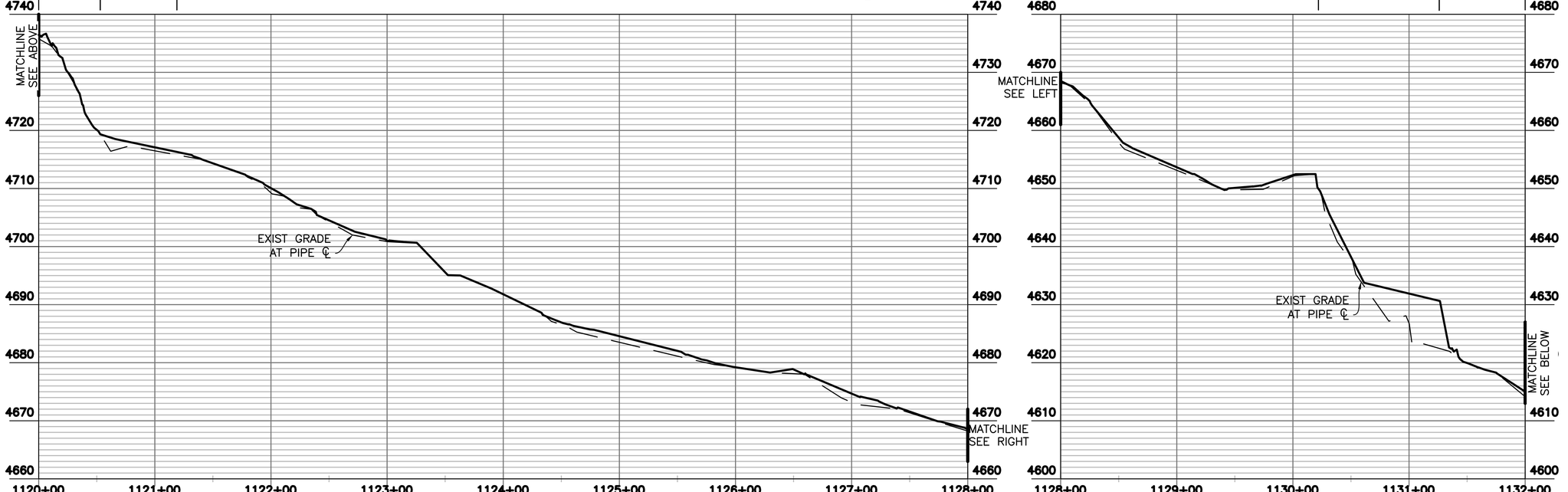
STA 1120+00 TO STA 1120+53 AT GRADE INSTALLATION — 2 / 111C501

STA 1120+53 TO STA 1121+19 ELEVATED INSTALLATION — 3 / 4 / 6 (111C501)

STA 1121+19 TO STA 1130+22 AT GRADE INSTALLATION — 2 / 111C501

STA 1130+22 TO STA 1131+26 ELEVATED INSTALLATION — 4 / 111C501

STA 1131+26 TO STA 1132+00 ELEVATED INSTALLATION — 6 / 111C501



GENERAL NOTES:

A. EXIST MV CABLE TO BE REMOVED AND DISPOSED WHERE ACCESSIBLE WITHOUT EXCAVATION. ALL BURIED CABLE (SOIL OR ROCKFALL) TO BE CUT AS CLOSE AS POSSIBLE TO BURIED AREA AND ABANDONED IN PLACE.

B. CONTRACTOR TO RESTORE DISTURBED AREAS TO PRECONSTRUCTION CONDITIONS.

C. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY GROUND DATA IS SHOWN TO PROVIDE A VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

D. EXIST ROARING SPRINGS PUMPHOUSE TO MANZANITA ALIGNMENT IS APPROXIMATE. MINOR DEVIATIONS IN THE NEW UTILITY ALIGNMENT ARE ANTICIPATED TO FACILITATE CONSTRUCTION. SEE SPECIFICATIONS FOR ALIGNMENT AND PROFILE REQUIREMENTS.

E. CONTRACTOR TO USE 6–INCH HDPE PIPE AS–NEEDED FOR SACRIFICIAL EXTERIOR SLEEVE DURING MV CABLE INSTALLATION TO PREVENT CABLE INSULATION DAMAGE DURING INSTALL.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
1120+00	1132+00	5 FT	2 FT

KEY NOTES: (X)

1. DRAINAGE CROSSINGS. PLACE MV CABLE INTO RIGID CONDUIT AND PLACE ADJACENT TO WATERLINE ON EXIST ROCK AND METAL PEDESTALS PER DETAILS 3 & 4 SHEET C501. EXTEND RIGID CONDUIT AT LEAST 20 LF ONTO AT GRADE PLACEMENT.
2. MV CABLE SURFACE SPLICE PER DETAIL 3 SHEET 111E501.
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50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 75014 — TOU VANG — State of ARIZONA, U.S.A.

Registered Professional Engineer (Civil) — CERTIFICATE NO. 76139 — NICHOLAS HUMPHREY — 10-9-2026 — Not Good ..... — ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. WALCHUK DRAWN: N. LYNCH TECH. REVIEW: S. EVERETT DATE: JUNE 2026	SUB SHEET NO. 111 C103	TITLE OF SHEET POWER DISTRIBUTION PLAN AND PROFILE STA 1120+00 TO 1132+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 5 OF 8

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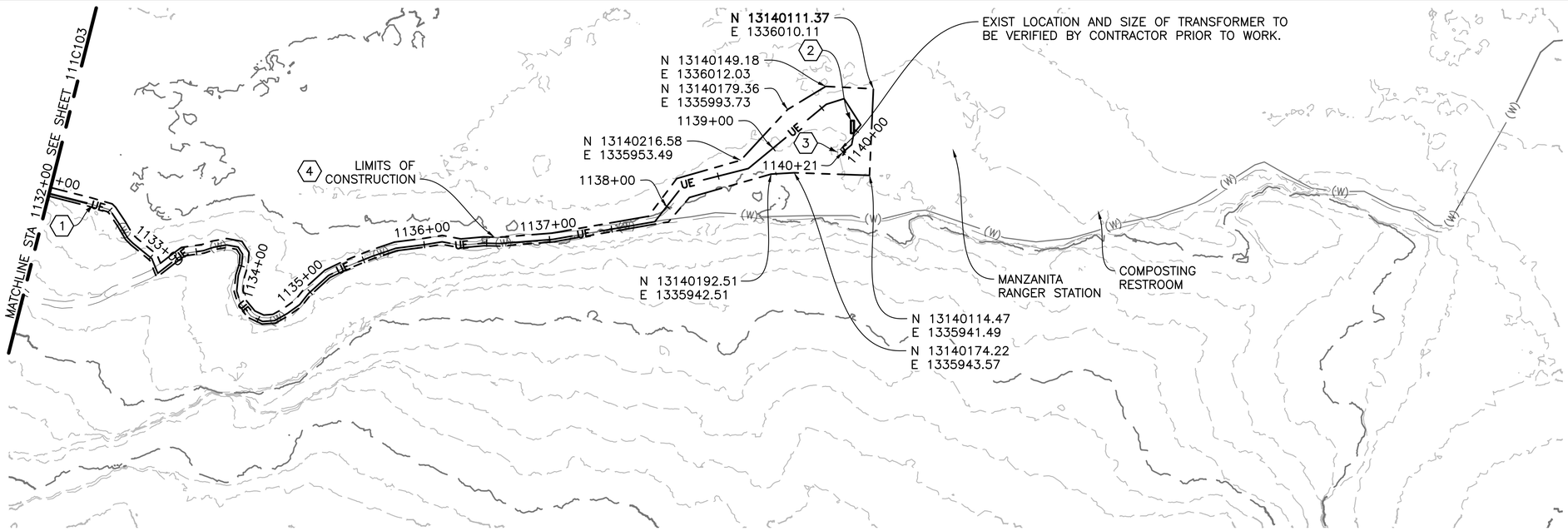
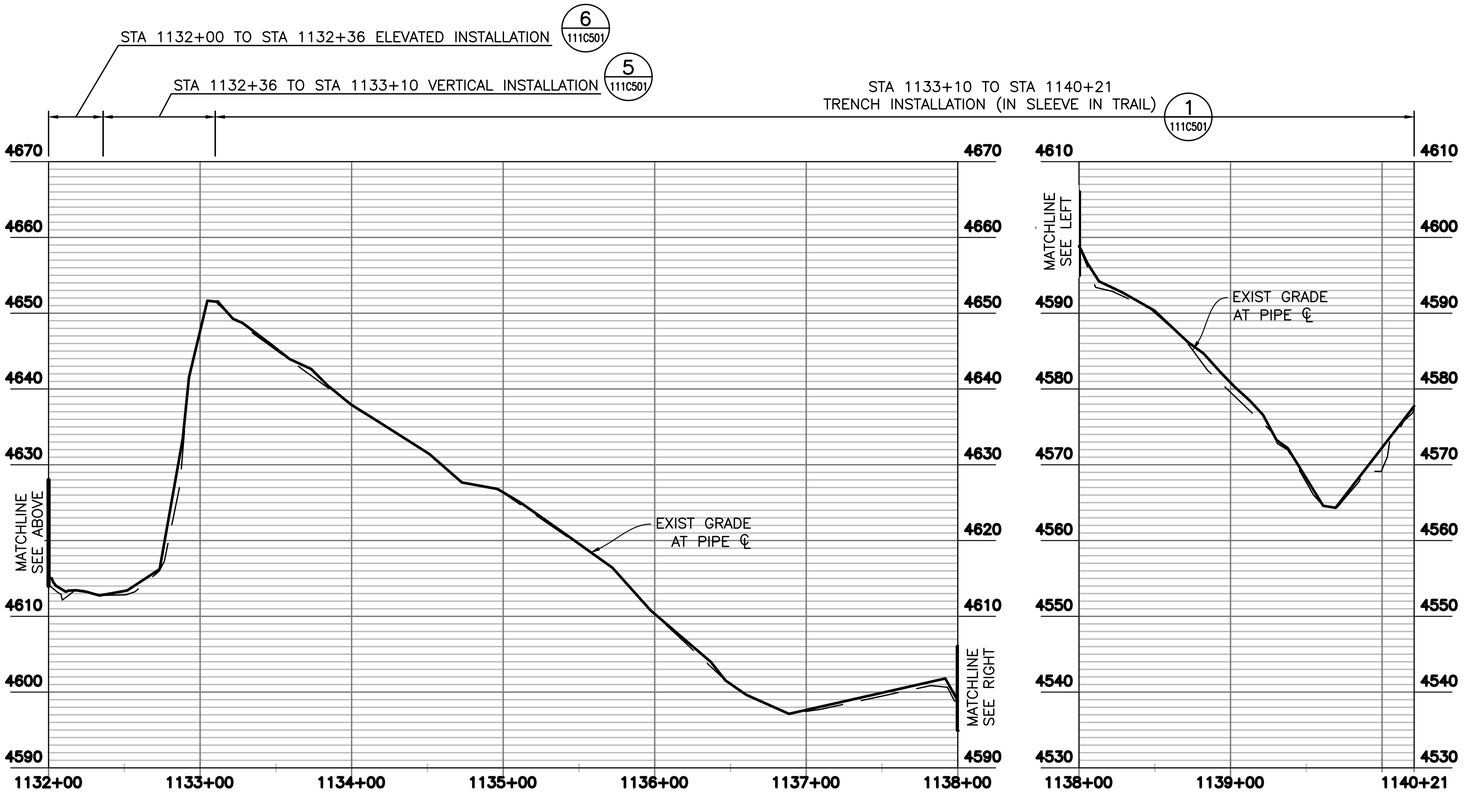




GENERAL NOTES:

A. EXIST MV CABLE TO BE REMOVED AND DISPOSED WHERE ACCESSIBLE WITHOUT EXCAVATION. ALL BURIED CABLE (SOIL OR ROCKFALL) TO BE CUT AS CLOSE AS POSSIBLE TO BURIED AREA AND ABANDONED IN PLACE.

B. CONTRACTOR TO RESTORE DISTURBED AREAS TO PRECONSTRUCTION CONDITIONS.

C. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY. GROUND DATA IS SHOWN TO PROVIDE VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

D. EXIST ROARING SPRINGS PUMPHOUSE TO MANZANITA ALIGNMENT IS APPROXIMATE. MINOR DEVIATIONS IN THE NEW UTILITY ALIGNMENT ARE ANTICIPATED TO FACILITATE CONSTRUCTION. SEE SPECIFICATIONS FOR ALIGNMENT AND PROFILE REQUIREMENTS.

E. CONTRACTOR TO USE 6–INCH HDPE PIPE AS–NEEDED FOR SACRIFICIAL EXTERIOR SLEEVE DURING MV CABLE INSTALLATION TO PREVENT CABLE INSULATION DAMAGE DURING INSTALL.

KEY NOTES: (X)

1. MV CABLE SURFACE SPLICE PER DETAIL 3 SHEET 111E501.
2. INSTALL 3–PHASE JUNCTION ENCLOSURE WITH 200AMP JUNCTION BARS. SEE SHEET 008E509 FOR TYPICAL DETAILS. TERMINATE MV CABLE WITH 200AMP LOAD BREAK ELBOWS. INSTALL 1–PHASE MV CABLE IN (2) 3" PVC CONDUIT FROM JUNCTION ENCLOSURE TO EXIST 1–PHASE PADMOUNT TRANSFORMER. LEAVE ONE CONDUIT AS A SPARE.
3. EXIST 1–PHASE PADMOUNT SERVICE TRANSFORMER. REMOVE EXIST MV CABLE TERMINATIONS. INTERCEPT EXIST CONDUIT AND TERMINATE 1–PHASE NON–ARMORED MV CABLE FROM JUNCTION ENCLOSURE WITH 200AMP LOAD BREAK ELBOWS. INSTALL SURGE ARRESTER ELBOW ON SPARE TRANSFORMER BUSHING.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
1132+00	1137+85	5 FT	2 FT
1137+85	1138+75	7 FT	10 FT
1138+75	1140+21	SEE POINTS ON PLAN VIEW	

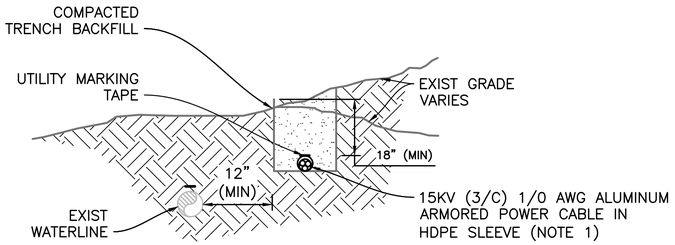
50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF ARIZONA U.S.A.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 10-9-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. WALCHUK DRAWN: N. LYNCH TECH. REVIEW: S. EVERETT DATE: JUNE 2026	SUB SHEET NO. 111 C104	TITLE OF SHEET **POWER DISTRIBUTION** PLAN AND PROFILE STA 1132+00 TO 1142+81 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 6 OF 8

10/12/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\111-C102.dwg

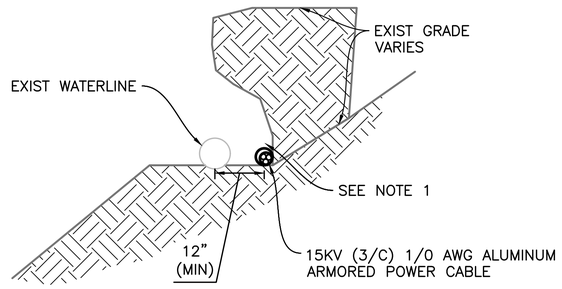


NOTES:

1. HDPE SLEEVE TO PROVIDE PROTECTION TO ARMORED CABLE DURING TRANSPORT, INSTALL, AND ANY FUTURE EXCAVATION OF WATERLINE.

1 / 111C501 TRENCH INSTALLATION (IN SLEEVE)

NOT TO SCALE

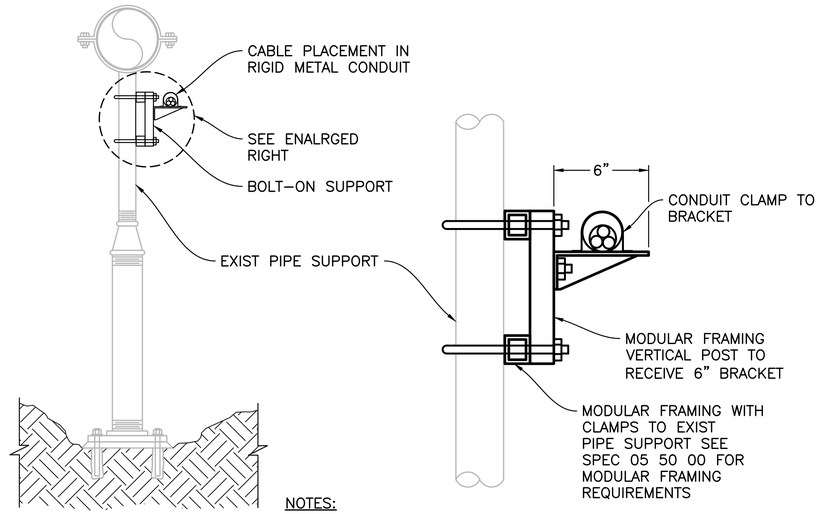


NOTES:

1. REMOVE LOOSE ROCKFALL/DEBRIS WHERE POSSIBLE TO ALLOW CABLE INSTALL ADJACENT TO WATERLINE. PLACE CABLE UNDER ANY ENCOUNTERED CABLE ANCHORS, AND PROTECT WATERLINE FROM DAMAGE AT ALL TIMES.
2. PLACE ARMORED POWER CABLE IN RIGID METAL CONDUIT FOR PROTECTION FROM ACCIDENTAL PHYSICAL DAMAGE.

2 / 111C501 AT GRADE INSTALLATION (UPSLOPE)

NOT TO SCALE

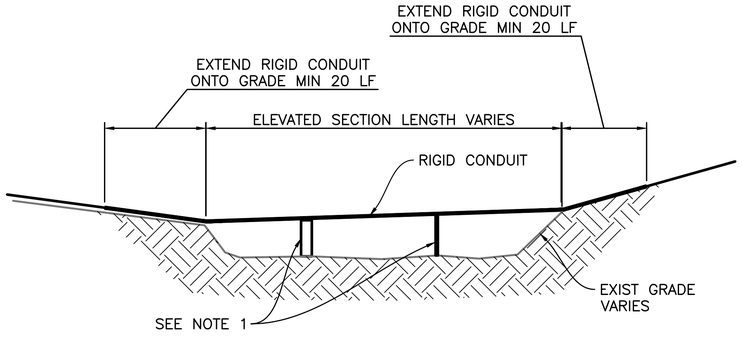


NOTES:

1. FOR ELEVATED INSTALLATION, CONTRACTOR TO SELECT EITHER DETAIL 3/111C501 OR 6/111C501 BASED ON EQUIPMENT ACCESS AND MATERIALS AT SUPPORT LOCATION.

3 / 111C501 EXIST PIPE SUPPORT MODIFICATION

NOT TO SCALE

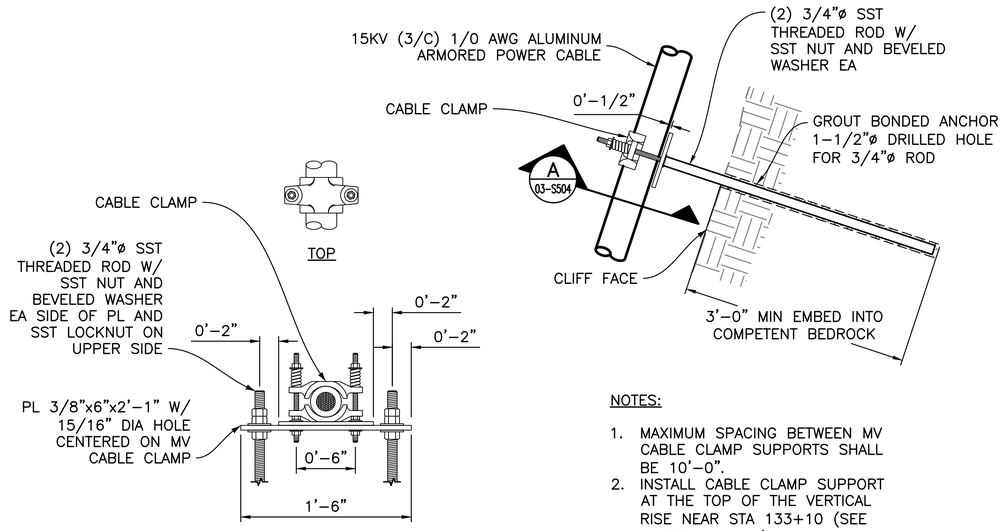


NOTES:

1. RIGID METAL CONDUIT TO HAVE SUPPORT EVERY 10–FT MIN.
2. PROPOSED PIPE SUPPORT TO BE INSTALLED BETWEEN EXIST PIPE SUPPORTS TO ACHIEVE 10–FT SUPPORT BETWEEN THE EXIST PIPE SUPPORTS. SEE VOLUME 0, SHEET 005D501 DETAIL 1, FLOOR PIPE SUPPORT.

4 / 111C501 RIGID CONDUIT SLEEVE

NOT TO SCALE

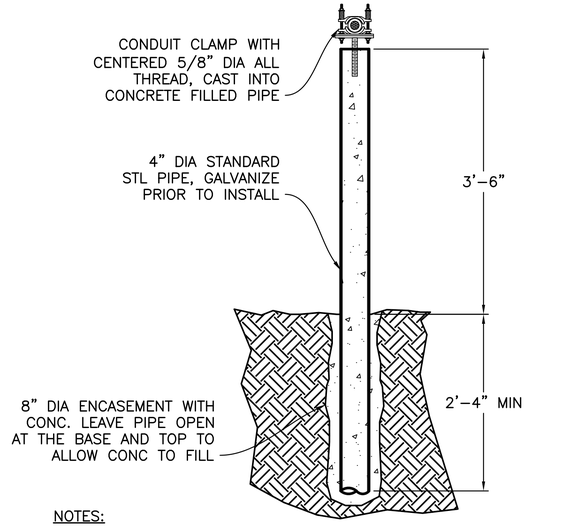


NOTES:

1. MAXIMUM SPACING BETWEEN MV CABLE CLAMP SUPPORTS SHALL BE 10'–0".
2. INSTALL CABLE CLAMP SUPPORT AT THE TOP OF THE VERTICAL RISE NEAR STA 133+10 (SEE SHEET 111C104)

5 / 111C501 VERTICAL CLIFF FACE INSTALLATION

NOT TO SCALE



NOTES:

1. FOR ELEVATED INSTALLATION, CONTRACTOR TO SELECT EITHER DETAIL 3/111C501 OR 6/111C501 BASED ON EQUIPMENT ACCESS AND MATERIALS AT SUPPORT LOCATION.

6 / 111C501 PROPOSED BOLLARD PIPE SUPPORT INSTALLATION

NOT TO SCALE

Registered Professional Engineer — Certificate No. 22198 — Nicholas D. LaFronz — Date Signed 8-1-2026 — Arizona, U.S.A.

Registered Professional Engineer — Certificate No. 85509 — Ian Coronado — Date Signed 8/1/2026 — Arizona, U.S.A.

Registered Professional Engineer (Civil) — Certificate No. 76139 — Nicholas Humphrey — 8-1-2026 — Date Signed — Arizona, U.S.A. — Nicholas Humphrey

Registered Professional Engineer — Certificate No. 75014 — Tou Vang — Date Signed

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. WALCHUK CADD N. LYNCH TECH. REVIEW: S. EVERETT DATE: AUG 2026	SUB SHEET NO. 111 C501	TITLE OF SHEET POWER DISTRIBUTION INSTALLATION DETAILS UNDERGROUND NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 8

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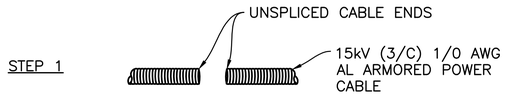
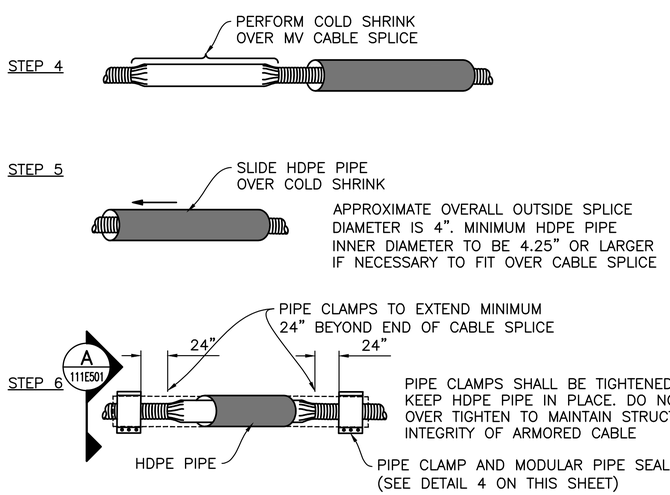
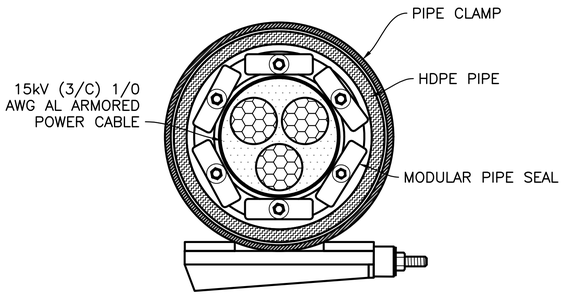




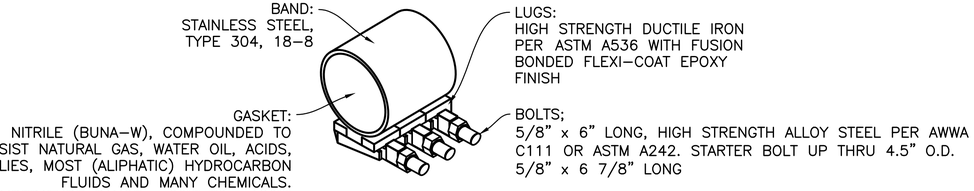


A 111E501 CROSS SECTION VIEW OF PIPE CLAMP

NOT TO SCALE

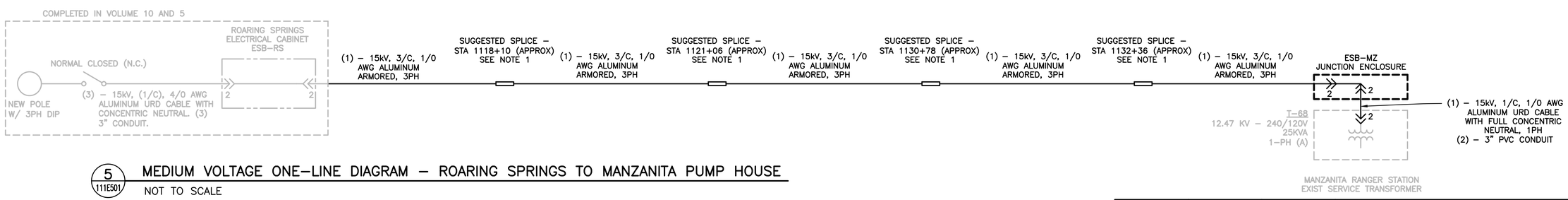
3 111E501 CONDUCTOR SPLICE FOR EXPOSED LOCATIONS

NOT TO SCALE



4 111E501 PIPE CLAMP

NOT TO SCALE



5 111E501 MEDIUM VOLTAGE ONE–LINE DIAGRAM – ROARING SPRINGS TO MANZANITA PUMP HOUSE

NOT TO SCALE

LEGEND:

2 200AMP ELBOW

TRANSFORMER

NOTE:

1. SPLICE PER DETAIL 3 THIS SHEET. LOCATIONS IDENTIFIED IN PLANS PROVIDED AS SUGGESTION ONLY. ACTUAL PLACEMENT TO BE DETERMINED IN FIELD BY CONTRACTOR.

REGISTERED PROFESSIONAL ENGINEER COLORADO CERTIFICATE NO. 75014 TOU VANG 2026

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: C. WALCHUK CADD N. LYNCH TECH. REVIEW: S. EVERETT DATE: JUNE 2026	SUB SHEET NO. 111 E501	TITLE OF SHEET POWER DISTRIBUTION VAULT AND SPLICE DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 8

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